

TECHBIG SOLUTIONS

Ambattur, Chennai

Microsoft Power BI – Complete E2E Training Syllabus

Data Analytics | Power Query | Data Modeling | DAX | Power BI Service

Program Objective: Build practical end-to-end Power BI capability covering data extraction, Power Query, M, semantic modeling, DAX, visualization, Power BI Service, security, governance, deployment, performance optimization, Fabric integration and real-world business dashboards.

Reference basis: Microsoft Learn and Power BI documentation covering data transformation, semantic modeling, DAX, Power BI Service, refresh, security, governance and end-to-end report deployment.

Program	Microsoft Power BI – Complete E2E
Level	Beginner → Advanced Data Analyst / BI Developer / Power BI Consultant
Core Areas	Power Query M Data Modeling DAX Visualization Service
Advanced Areas	RLS/OLS Gateway Refresh Governance Deployment Performance Fabric
Suggested Duration	100–140 hours, customizable by batch and project depth

1. Power BI Overview & Business Intelligence

- Power BI ecosystem: Desktop, Power BI Service, Mobile, Report Server and Embedded
- BI concepts, self-service analytics and enterprise reporting
- Reports, dashboards, semantic models, apps and workspaces
- End-to-end analytics lifecycle from raw data to business insight
- Power BI licensing and deployment concepts

2. Power BI Desktop – Complete Fundamentals

- Installation and interface
- Report, Data and Model views
- PBIX/PBIP project concepts
- Fields, filters, visualizations and formatting
- Themes, bookmarks and navigation
- Basic report authoring workflow

3. Data Sources & Connectivity

- Excel, CSV, TXT, JSON and XML
- SQL Server and relational databases
- Azure SQL and cloud data sources
- SharePoint and OneDrive
- Web and API sources
- Folder and multiple-file ingestion
- Import, DirectQuery and Direct Lake concepts
- Connection credentials and privacy

4. Power Query – Data Extraction & Transformation

- Power Query Editor
- Data profiling and quality

- Filtering, sorting and removing duplicates
- Data types and error handling
- Split, merge, append and group operations
- Pivot/unpivot and transpose
- Replace values and conditional columns
- Custom columns and parameters
- Query dependencies and folding concepts

5. Power Query M Language

- M language fundamentals
- Functions, variables and expressions
- Custom functions
- Lists, records and tables
- Error handling
- Reusable transformation logic
- Performance-aware M development

6. Data Cleaning & ETL Best Practices

- Source validation and cleansing
- Null and missing-value handling
- Data standardization
- Date and text cleansing
- Reference vs duplicate queries
- Staging and transformation patterns
- Incremental and refresh-friendly design

7. Data Modeling Fundamentals

- Tables, columns, measures and hierarchies
- Relationships and cardinality
- Active/inactive relationships
- Single vs bidirectional filtering
- Star schema and dimensional modeling
- Fact and dimension tables
- Many-to-many relationships and bridge tables

8. Semantic Models & Model Design

- Configure tables and columns
- Keys and data types
- Hierarchies and display folders
- Measures and calculation design
- Parameters and field parameters
- Date tables and model organization
- Reusable enterprise semantic models

9. DAX Fundamentals

- DAX syntax and data types
- Measures vs calculated columns vs calculated tables
- Basic aggregation functions
- CALCULATE and filter context
- Variables and readable DAX
- IF, SWITCH and logical expressions
- Common text, numeric and date functions

10. Advanced DAX

- Row context and filter context
- Context transition
- CALCULATE/CALCULATETABLE
- FILTER and iterator functions
- ALL, ALLEXCEPT, ALLSELECTED and REMOVEFILTERS
- RELATED and RELATEDTABLE
- SUMX, AVERAGEX, COUNTX and other iterators
- Virtual tables and advanced calculation patterns

11. DAX Time Intelligence

- Calendar/date table
- YTD, MTD, QTD and rolling periods
- Previous period and same-period-last-year analysis
- Year-over-year and month-over-month growth
- Running totals and moving averages
- Fiscal calendars
- Period comparison and trend measures

12. Advanced Analytics & Calculation Patterns

- Dynamic titles and dynamic measures
- Ranking and Top N
- Percent-of-total calculations
- Pareto analysis
- Variance and contribution analysis
- What-if parameters
- Dynamic segmentation and KPI logic

13. Data Visualization & Charts

- Bar, column, line and area charts
- Pie/donut and treemap
- Scatter, waterfall, funnel and gauge
- Cards, KPI and multi-row cards
- Tables and matrix
- Maps and geographic visuals
- Conditional formatting and data labels

14. Interactive Report Design

- Slicers and filter panes
- Drill-down and drill-up
- Drill-through pages
- Tooltips and report-page tooltips
- Bookmarks and buttons
- Page navigation
- Sync slicers
- Accessibility and user-friendly layouts

15. Advanced Visuals & Storytelling

- Decomposition tree
- Key influencers
- Smart narrative and analytical visuals
- Small multiples
- Reference lines and analytics pane
- Custom visuals
- Executive dashboard storytelling
- Design standards and UX

16. Power BI Service

- Power BI Service navigation
- Workspaces and workspace roles
- Publishing from Desktop
- Reports and semantic models
- Dashboards and tiles
- Apps and audiences
- Sharing and collaboration
- Subscriptions and alerts

17. Semantic Model Management & Refresh

- Dataset/semantic-model settings
- Scheduled refresh
- Refresh history and failure analysis
- On-premises data gateway
- Data-source credentials
- Incremental refresh
- Parameters and deployment-ready refresh design

18. Power BI Security

- Workspace security
- App permissions
- Row-level security (RLS)
- Static and dynamic RLS

- Object-level security (OLS)
- User and group-based access
- SSO and source security concepts
- Security testing and validation

19. Power BI Governance & Administration

- Tenant and capacity concepts
- Governance standards
- Endorsement and certification
- Sensitivity labels and information protection
- Usage metrics and lineage
- Impact analysis
- Deployment and content lifecycle
- Audit and compliance concepts

20. Power BI Deployment & DevOps

- Development, test and production environments
- Deployment pipelines
- PBIP and source-control concepts
- Version management
- Release strategy
- Parameters and environment-specific configuration
- Rollback and production support

21. Performance Optimization

- Model size and cardinality
- Star schema optimization
- Power Query query folding
- DAX performance
- Storage modes and composite models
- Aggregation concepts
- Performance Analyzer
- Refresh and report optimization

22. DirectQuery, Composite Models & Advanced Connectivity

- DirectQuery architecture
- Import vs DirectQuery trade-offs
- Composite models
- Dual storage mode
- Aggregations
- Cross-source modeling
- Real-time and near-real-time scenarios

23. Paginated Reports & Report Builder

- Paginated report concepts

- Report Builder interface
- Tables, matrices and charts
- Parameters and filters
- Expressions and formatting
- Pixel-perfect operational reporting
- Publishing and distribution

24. Power BI Mobile & Embedded

- Mobile layouts and consumption
- Mobile app security
- Power BI Embedded overview
- Embedding concepts
- Service principals and application integration
- Embedded analytics use cases

25. Power BI with Microsoft Fabric

- Fabric and Power BI relationship
- OneLake concepts
- Lakehouse and warehouse connectivity
- Semantic models in Fabric
- Direct Lake overview
- Fabric-based enterprise analytics
- Data engineering/warehouse integration concepts

26. AI & Copilot in Power BI

- AI-assisted report authoring concepts
- Copilot capabilities and prerequisites
- Natural-language exploration
- AI-assisted summaries and insights
- Responsible AI and governance
- Validation of AI-generated results

27. Excel & Office Integration

- Analyze in Excel
- Export data and reports
- PowerPoint integration
- Excel/Power BI semantic-model usage
- Office sharing and collaboration scenarios

28. SQL for Power BI Analysts

- SELECT, WHERE, GROUP BY and ORDER BY
- JOINS and subqueries
- Aggregations and window functions
- CTEs and views
- SQL for source-side transformations

- Query optimization basics

29. Real-Time E2E Projects

- Sales dashboard: SQL/Excel → Power Query → model → DAX → executive report
- Finance dashboard: P&L, budget vs actual, variance and KPI analysis
- HR analytics: headcount, attrition, hiring and department analysis
- Inventory dashboard: stock, aging, movement and reorder analysis
- Procurement dashboard: supplier spend, PO and delivery performance
- Production dashboard: output, downtime, quality and efficiency
- Customer analytics: revenue, retention, segmentation and profitability
- Power BI + SAP analytics: ECC/S4HANA data extraction and management reporting
- Enterprise workspace: RLS, refresh, gateway, app distribution and governance
- End-to-end deployment: development → test → production with monitoring

30. Troubleshooting & Production Support

- Refresh failures
- Gateway connectivity issues
- Data-source credential errors
- Relationship/model issues
- DAX calculation discrepancies
- Slow report performance
- RLS access problems
- Publishing and workspace issues
- Visual rendering issues
- Production incident and root-cause analysis

31. Interview & Certification Preparation

- Power BI interview fundamentals
- Power Query and M questions
- DAX scenario-based questions
- Data modeling case studies
- Power BI Service administration questions
- RLS and governance scenarios
- Performance-tuning questions
- Real-time project explanation
- Microsoft Power BI Data Analyst certification preparation

Recommended Hands-on Deliverables

- Build a complete sales/finance dashboard from raw Excel or SQL data through Power Query, semantic model, DAX and published Power BI app.
- Create a star-schema model with relationships, measures, date intelligence and advanced DAX.
- Implement dynamic RLS, scheduled refresh, gateway connectivity and workspace/app security.
- Optimize a slow Power BI report using model design, query folding, DAX and Performance Analyzer.
- Develop an executive KPI dashboard with drill-through, bookmarks, tooltips, navigation and mobile layout.

- Complete a production deployment from development to test to production with governance and monitoring.

Training Outcome

Learners completing this program should be able to independently build and deploy Power BI solutions, transform and model data, write production-quality DAX, create executive dashboards, secure and refresh enterprise reports, troubleshoot production issues and participate in BI implementation projects.

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